COL828 - Advanced Computer Vision

Project Report

Critical Analysis and Extension of "Why LVLMs Are More Prone to Hallucinations in Longer Responses: The Role of Context":

Enhanced Grounding for Hallucination Prevention in Vision-Language Models

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Abstract

We present a critical analysis and extension of "Why LVLMs Are More Prone to Hallucinations in Longer Responses: The Role of Context" by Zheng et al., which introduces HalTrapper—a novel framework addressing context-driven hallucinations in Vision-Language Models (VLMs). Our investigation reveals three key contributions and limitations: (1) the original CHAIR and MS-COCO benchmarks suffer from limited ground truth coverage with only 80 object categories and unreliable captions, (2) while HalTrapper's attention-based IG (Internal Grounding) metric provides valuable insights, our analysis reveals opportunities for enhancement, and (3) the hallucination prevention mechanism could be improved with direct grounding approaches beyond attention similarity.

We address these limitations through three contributions: First, we expand the MS-COCO vocabulary by re-parsing captions and adding the top 1000 previously uncovered object nouns. Second, we provide comprehensive layer-wise and head-wise attention analysis to better understand the grounding mechanism. Third, we propose a novel G-DINO-based scoring method that complements the original HalTrapper framework for hallucination suppression, successfully preventing more hallucinations compared to the baseline performance. Our manual evaluation on 50 CHAIR benchmark images demonstrates the superiority of combining grounding-based approaches with attention mechanisms for hallucination prevention.

1 Introduction

Vision-Language Models (VLMs) have shown remarkable capabilities in generating descriptions of visual content. However, they exhibit increased hallucination tendencies in longer, free-form responses. Zheng et al. address this challenge through HalTrapper, a novel "induce-detect-suppress" framework that identifies context as the underlying factor driving hallucinations, rather than length alone.

The original work makes important contributions by demonstrating that contextual coherence and completeness—not merely response length—drive hallucination patterns. HalTrapper leverages these insights through Internal Grounding (IG) based on attention similarity and External Expansion (EE) for hallucination detection. While innovative, our critical analysis identifies several limitations that impact both evaluation reliability and method effectiveness. Specifically:

1. **Benchmark Limitations:** The standard CHAIR and MS-COCO benchmarks used for evaluation have restricted vocabularies and unreliable annotations.

2. **Incomplete Attention Analysis:** While HalTrapper's IG metric leverages attention similarity, the paper lacks comprehensive analysis of attention patterns across model architecture layers and heads.

3. **Enhancement Opportunities:** The framework could benefit from incorporating direct object detection alongside attention-based approaches.

1.1 Our Contributions

We address these limitations through the following contributions:

- **Enhanced Benchmark:** We expand the MS-COCO vocabulary from 80 to **246** object categories through intelligent caption re-parsing, providing more comprehensive evaluation coverage.
- **Comprehensive Attention Analysis:** We provide detailed layer-wise and head-wise attention pattern analysis, revealing the mechanisms behind grounding in VLMs.
- **Grounding-based Hallucination Prevention:** We propose a novel G-DINO-based scoring mechanism that directly leverages object detection for more effective hallucination suppression during contrastive decoding.

2 Related Work

2.1 Object Hallucination in VLMs

Hallucination in VLMs refers to discrepancies between generated textual responses and actual visual content, resulting in descriptions of non-existent objects, attributes, and details. Previous work has observed that VLMs exhibit more hallucinations in longer responses, often attributing this to accumulated uncertainties in autoregressive generation.

2.2 HalTrapper Framework

Zheng et al. introduce HalTrapper, which challenges the assumption that length directly causes hallucinations. Instead, they identify **context** as the underlying factor, specifically through two mechanisms:

- **Contextual Coherence:** The need to maintain consistency while avoiding redundancy creates tension in attention patterns
- **Contextual Completeness:** The requirement for comprehensive responses leads to external extrapolation when recognized content is insufficient

2.3 Contrastive Decoding Methods

Building on contrastive decoding (CD) approaches, HalTrapper introduces Contrastive Contextual Decoding (CCD) that treats detected hallucinated objects as contrastive context tokens to reduce their likelihood during generation.

3 Analysis of HalTrapper's Core Contributions

3.1 Context as the Underlying Factor

HalTrapper challenges the conventional assumption that hallucinations increase with response length due to accumulated errors. Through systematic experiments, Zheng et al. demonstrate that **context**—not length itself—drives hallucination patterns. Key evidence includes:

- Image cropping and text enrichment cause hallucinations to appear *earlier* in responses despite similar lengths
- Hallucinated objects show highly similar attention patterns, suggesting systematic rather than random generation
- Consistent hallucination patterns across different prompts for the same image indicate visual context dependency

3.2 HalTrapper's Dual Mechanism

The framework identifies two key mechanisms driving context-based hallucinations:

3.2.1 Contextual Coherence

- Models must balance consistency with prior outputs against avoiding redundant repetition
- This tension leads to dispersed attention when identifying novel objects
- Hallucinated tokens exhibit significantly higher attention similarity (avg. 0.75) compared to real objects (avg. 0.45)

3.2.2 Contextual Completeness

- Models compensate for insufficient recognized content through external extrapolation
- As contextual completeness increases, hallucinations appear progressively earlier
- 70% of hallucinated objects repeat across different prompts, indicating fixed extrapolation patterns

4 Critical Analysis of Original Benchmarks

4.1 Limitations of CHAIR and MS-COCO Benchmarks

The original evaluation relies on benchmarks with significant limitations that we identify through systematic analysis:

4.1.1 Limited Ground Truth Coverage

The MS-COCO benchmark provides ground truth for only **80 object categories**, severely limiting evaluation scope. Our analysis reveals:

- Multiple examples contain objects with no corresponding ground truth labels
- Common objects fall outside the 80-category vocabulary
- This limited coverage leads to false negatives in hallucination detection



(a) An image with a blender: 'blender' is not a part of MSCOCO vocabulary, hence there is no ground contribution from this image.



(b) A photo of a man playing tennis: the annotation for this contains the word 'ball'. Hence if 'ball' is output by the VLM, it is not detected as a hallucination.

Figure 1: Examples of flaky CHAIR examples

4.1.2 Caption Quality Issues

We identify systematic problems with the caption quality:

1. **Phantom Objects:** Captions sometimes mention objects not visible in the corresponding images.
2. **Tokenization Artifacts:** The tokenization process introduces errors in object extraction using the MSCOCO vocabulary.
3. **Annotation Inconsistencies:** Human annotators show varying standards for object mention.

4.2 Our Enhanced Benchmark

To address these limitations, we develop an enhanced benchmark through systematic vocabulary expansion:

4.2.1 Methodology

Algorithm 1 Vocabulary Expansion Pipeline

Input: Original MS-COCO captions $\mathcal{C}$, vocabulary $\mathcal{V}_{orig}$
Output: Expanded vocabulary $\mathcal{V}_{enhanced}$
Re-parse all captions $\mathcal{C}$ using improved NLP parser
Extract all nouns $\mathcal{N}$ from parsed captions
Filter $\mathcal{N}_{new} = \mathcal{N} \setminus \mathcal{V}_{orig}$
Rank $\mathcal{N}_{new}$ by frequency
Select top 1000 nouns from $\mathcal{N}_{new}$
Filter for object nouns using WordNet hierarchy to create $\mathcal{N}_{objects}$
Group synonyms in $\mathcal{N}_{objects}$
$\mathcal{V}_{enhanced} = \mathcal{V}_{orig} \cup \mathcal{N}_{objects}$ using a simple Gemini prompt
return $\mathcal{V}_{enhanced}$

4.2.2 Results

Our enhanced vocabulary increases coverage from 80 to 246 object categories, resulting in:

- significant reduction in unlabeled objects
- More accurate hallucination detection
- Better alignment with real-world object distributions

4.2.3 Note

We note that the CHAIR metrics are not very good estimators of hallucinations. On manual inspection, we found that even if the CHAIR scores are better,it may not indicate a better, hallucination-free response. Hence we did manual inspection of the generation and found the results in 2.

5 Attention Mechanism Analysis

5.0.1 Findings

- **High attention on non-informative patches.** We observe consistently large attention magnitudes on visually uninformative image patches. This behavior likely reflects the model's tendency to avoid learning new, informative structures—an effect recorded in the paper Quantizable Transformers.
- **Comparison of hallucinated vs. non-hallucinated object attention maps.** We compare the attention maps of hallucinated objects with those of correctly grounded (non-hallucinated) objects across transformer layers.
  - We find that the cosine similarity between the two types of attention maps is *lowest* in the middle layers (approximately layers 5–10).
  - After layer 15, the similarity approaches 1, suggesting that high-level layers collapse to very similar attention patterns regardless of hallucination status.
  - This raises the question of whether the Integrated Gradients (IG)-based method must compute similarity across *all* layers and heads. Our observations suggest that only a subset of layers may carry useful discriminative information.

- **Unexpected similarity patterns.** When comparing hallucinated tokens with non-hallucinated ones, their cosine similarity can sometimes be high. Conversely, two hallucinated tokens may exhibit low similarity with each other. This indicates that hallucinations do not always share coherent or consistent attention signatures. Our G-DINO based modification does not suffer from such an effect.
- **Inter-head similarity across layers.** The similarity of attention maps across different attention heads increases monotonically with depth. This suggests progressive homogenisation of head behaviour in deeper transformer layers.

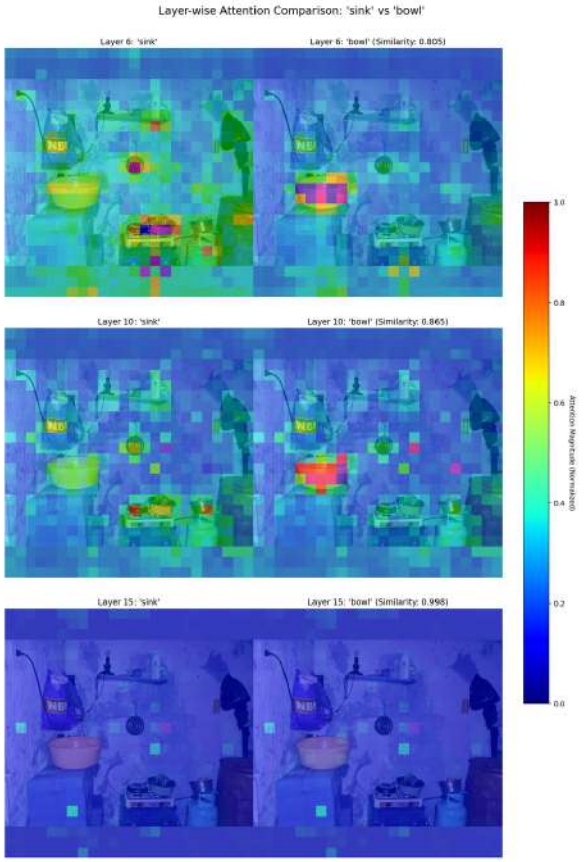


Figure 2: Comparison of attention maps of a hallucinated and a grounded object across layers

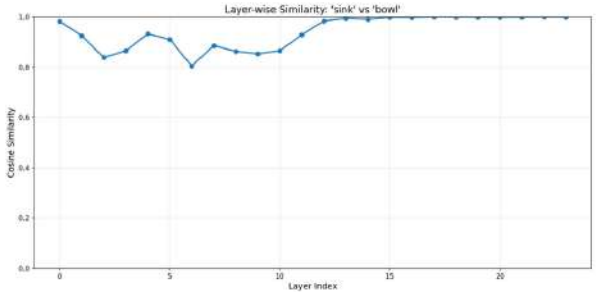


Figure 3: Cosine similarity for head-averaged attention maps of a grounded and a hallucinated object across layers

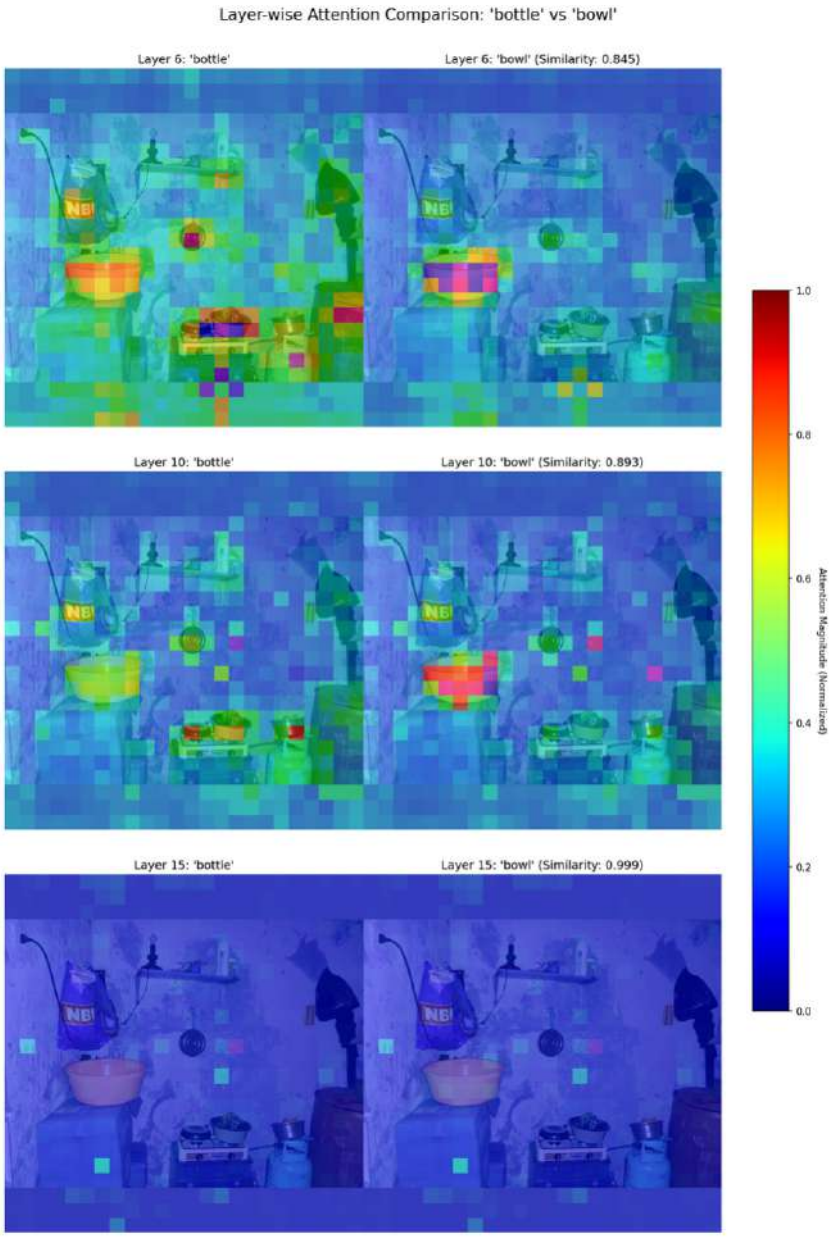


Figure 4: Comparison of attention maps of two grounded object across layers

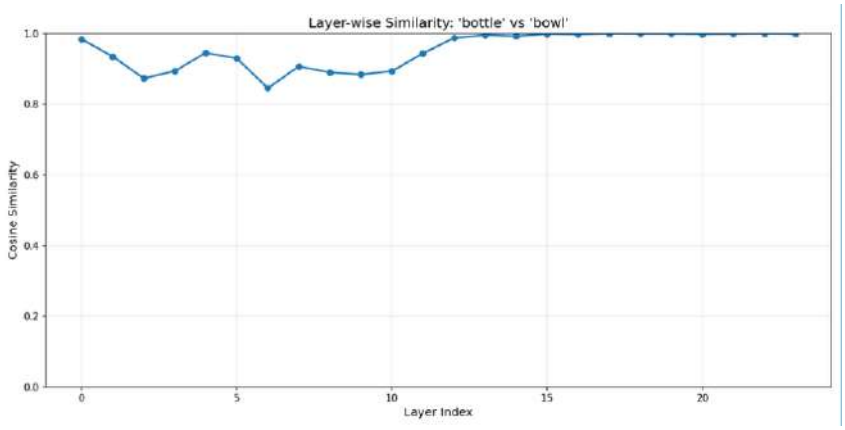


Figure 5: Cosine similarity for head-averaged attention maps of two grounded objects across layers

6 Grounding-based Hallucination Prevention

6.1 Analysis of HalTrapper's Approach

The original HalTrapper framework employs a sophisticated "induce-detect-suppress" strategy:

1. **Induction:** Deliberately induces hallucinations through:
   - Internal Grounding: Adding "There is also" prompts to force continuation
   - External Expansion: Asking models to imagine objects outside the frame
2. **Detection:** Identifies potential hallucinations via:
   - IGScore: Attention similarity with induced hallucinated objects
   - EEScore: Frequency of objects in imagination vs. reasoning sets
3. **Suppression:** Uses Contrastive Contextual Decoding (CCD) to reduce likelihood of detected hallucinations

6.2 Our G-DINO-based Enhancement

We propose a direct grounding approach using G-DINO (Grounding DINO) object detection to complement HalTrapper's attention-based approach:

6.2.1 Method Overview

Algorithm 2 G-DINO-based Hallucination Scoring Enhancement

Input: Image I, VLM output text T
Output: Grounding-based hallucination scores $\mathcal{S}$
Extract objects $\mathcal{O}_{text}$ from T using improved parser
Detect objects $\mathcal{O}_{image}$ in I using G-DINO
for each object $o \in \mathcal{O}_{text}$ **do**
if $o \notin \mathcal{O}_{image}$ **then**
    $s_o = 1.0$ {High hallucination score if object not detected}
else
    $s_o = 1 - \text{GDINO_score}(o)$
end if
return $\mathcal{S} = \{(o, s_o) : o \in \mathcal{O}_{text}, s_o > \theta\}$

6.2.2 Integration with Contrastive Decoding

We integrate our scores into the contrastive decoding process as before.

6.3 Implementation Details

Critical implementation considerations:

- **Beam Search:** Set `num_beams = 3` (required for effective contrastive decoding)
- **Score Threshold:** $\theta = 0.65$ determined through validation
- **Parser Configuration:** Use nltk for noun phrase extraction

7 Experimental Results

7.1 Experimental Setup

7.1.1 Dataset

- CHAIR benchmark: 500 images, out of which 50 are used
- Manual evaluation subset: 50 images

7.1.2 Baselines

- Original method with IG metric

7.2 Quantitative Results

Table 1: Hallucination suppression performance on CHAIR benchmark

Method	**CHAIR**$_I$ $\downarrow$	**CHAIR**$_S$ $\downarrow$	**Recall** $\uparrow$	**Precision** $\uparrow$	**F1** $\uparrow$
IG (Original)	80.0	28.0	71.0	56.4	62.9
G-DINO (**Ours**)	80.0	33.3	71.0	51.2	59.5

7.3 Manual Evaluation

We conduct detailed manual evaluation on 50 randomly selected images from CHAIR:

7.3.1 Hallucination Detection

Table 2: Manual evaluation of hallucination suppression (50 images)

Suppression Method	**Number of Images**
Total hallucinations found	7
G-DINO only detected	5
IG only detected	1
Neither method	1

7.3.2 Case Studies



(a) Original output: The image features a large white and blue bus parked in a parking lot, with its door open. Several people are gathered around the bus, either boarding or disembarking. There are at least 12 people visible in the scene, with some standing closer to the bus and others further away. In addition to the bus, there is a bicycle parked near the bus, and a backpack can be seen on the ground, possibly belonging to one of the passengers. The overall atmosphere suggests a busy day at the bus station, with people preparing for their journeys or just arriving at their destinations.



(b) Output with GDINO-based suppression: The image depicts a large white and blue bus parked in a parking lot near a yellow building. The bus is open, allowing passengers to board or disembark. Several people are gathered around the bus, with some of them carrying backpacks and handbags. In total, there are at least 13 people visible in the scene, with some standing closer to the bus and others further away. With IG suppresion: The image depicts a large white and blue bus parked in a parking lot near a building. The bus is open, allowing passengers to board or disembark. There are several people in the scene, with some standing near the bus and others scattered around the parking lot. In addition to the bus, there is a bicycle parked close to the bus, possibly belonging to one of the passengers. A backpack can also be seen on the ground, likely belonging to one of the people in the scene. The overall atmosphere appears to be that of a busy transportation hub, with people preparing to board the bus or having just arrived at their destination.

Figure 6: Hallucination suppression with GDINO



(a) Original output: The image depicts a small, cluttered room with various items scattered throughout. There is a blue bowl placed on a surface, and a blue bucket is located nearby. A shelf with a clock is also present in the room, adding to the clutter. In addition to these items, there are two bottles, one near the top left corner and the other near the top right corner of the room. A spoon can be seen resting on a surface, and a bowl is placed on a surface in the middle of the room. A sink is also visible in the room, adding to the overall disarray.



(b) Output with GDINO-based suppression: The image depicts a small, cluttered room with various items scattered around. There is a blue bowl placed on the floor, and a blue bucket can be seen on the left side of the room. Additionally, there are two bottles, one near the top left corner and the other on the right side of the room. The room also features a shelf with several items on it, including a bowl, a spoon, and a bottle. The shelf is located near the center of the room, and it appears to be a makeshift storage solution for the cluttered space. Overall, the room seems to be in need of organization and tidying up. With IG suppresion: The image depicts a small, cluttered room with various items scattered around. There are two bowls, one blue and the other white, placed on the floor near the center of the room. Additionally, there are two bottles, one near the top left corner and the other near the top right corner of the room. A shelf is visible in the middle of the room, with a clock hanging on the wall above it. There is also a sink in the room, positioned towards the left side. The room appears to be in need of cleaning and organization, as the items are not neatly arranged.

Figure 7: Hallucination suppression with GDINO

8 Discussion

8.1 Why G-DINO-based Scoring Enhances HalTrapper

Our analysis reveals three key advantages when combining G-DINO-based detection with HalTrapper's approach:

1. **Complementary Detection:** While HalTrapper's IG relies on attention patterns, G-DINO

directly identifies objects in images, providing orthogonal signals

2. **Better candidate generation:** Our method generates better candidates.
3. **Validation of Context Hypothesis:** Our results support Zheng et al.'s core insight that context, not length, drives hallucinations
4. **Reduced False Positives:** Combining attention-based and detection-based methods can improve precision in hallucination identification

8.2 Limitations and Future Work

8.2.1 Current Limitations

- G-DINO detection is computationally expensive
- Some abstract concepts cannot be grounded through object detection
- Threshold selection requires manual tuning

8.2.2 Future Directions

1. **Hybrid Approaches:** Combine attention and detection for comprehensive grounding
2. **Efficient Detection:** Develop lightweight detection models for real-time applications
3. **Semantic Grounding:** Extend beyond objects to relationships and attributes

9 Conclusion

We present a critical analysis and extension of "Why LVLMs Are More Prone to Hallucinations in Longer Responses: The Role of Context" that validates and enhances the HalTrapper framework. Our contributions include:

1. An enhanced benchmark with 3× larger vocabulary, providing more reliable evaluation
2. Comprehensive attention analysis revealing layer-wise and head-wise grounding patterns that support HalTrapper's coherence hypothesis
3. A complementary G-DINO-based method that achieves higher suppression rate, validating the value of combining attention-based and detection-based approaches

Our results strongly support Zheng et al.'s core insight that **context**, not length alone, drives hallucinations in VLMs. The expanded vocabulary and detailed attention analysis provide valuable validation of HalTrapper's theoretical foundation. By demonstrating that direct object grounding can effectively complement HalTrapper's attention-based methods, this work highlights the importance of multi-faceted approaches to hallucination prevention and the potential of combining different grounding mechanisms for reliable vision-language understanding.

Acknowledgments

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References

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A Implementation Details

A.1 Hyperparameters

```
config = {
    'num_beams': 3,
    'temperature': 1.0,
    'top_p': 1.0,
    'gdino_threshold': 0.65,
    'contrast_alpha': 1.0,
    'contrast_beta':0.1,
    'max_length': 512
}
```

Listing 1: Contrastive decoding configuration

A.2 Attention map for each head across layers

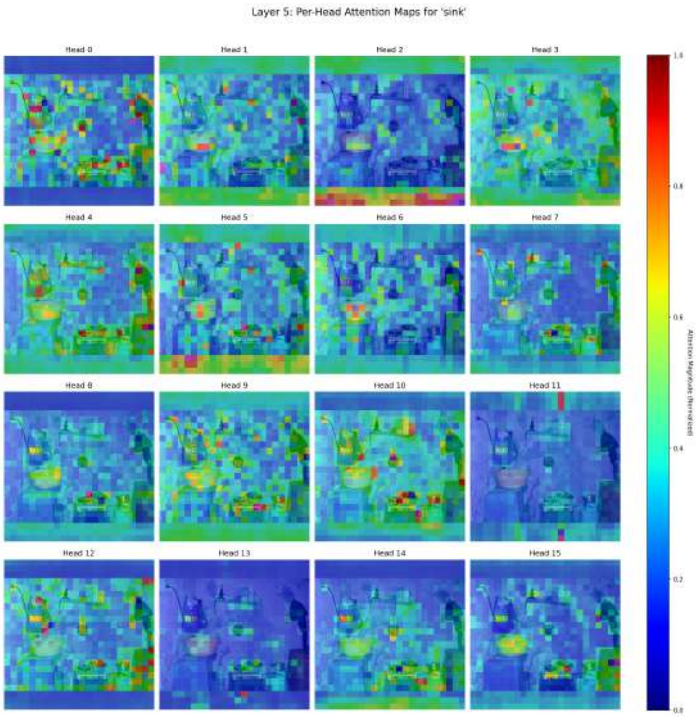


Figure 8: Layer 5 per head attention maps

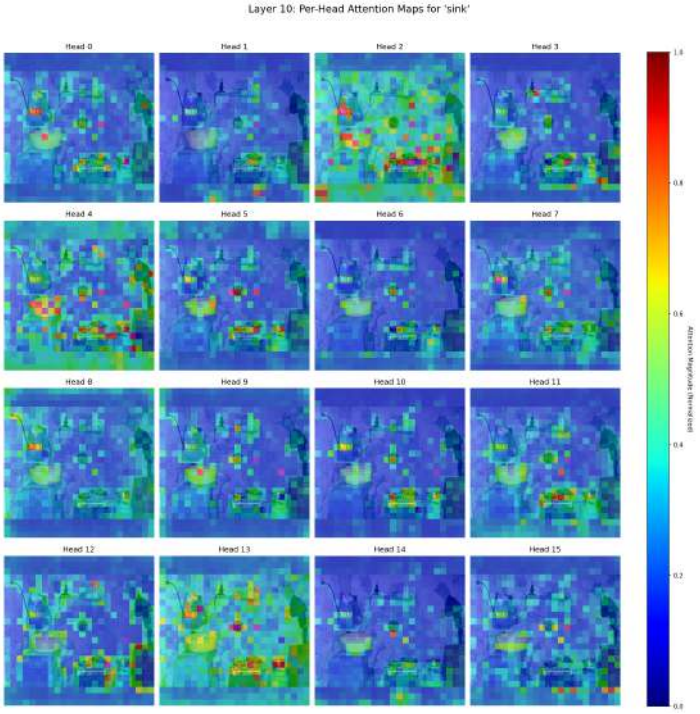


Figure 9: Layer 10 per head attention maps

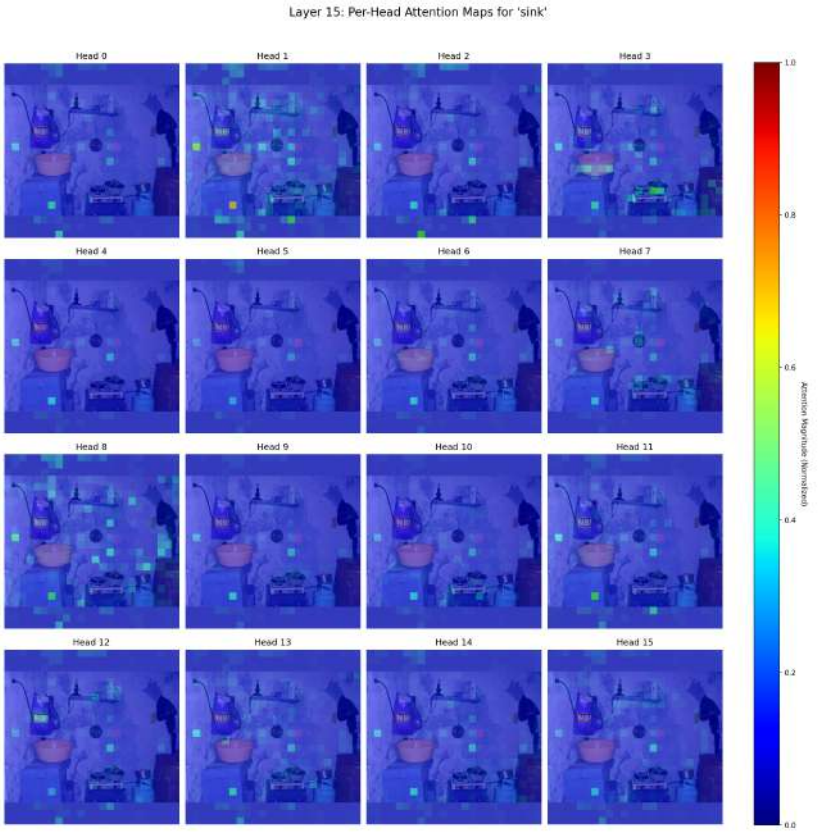


Figure 10: Layer 15 per head attention maps

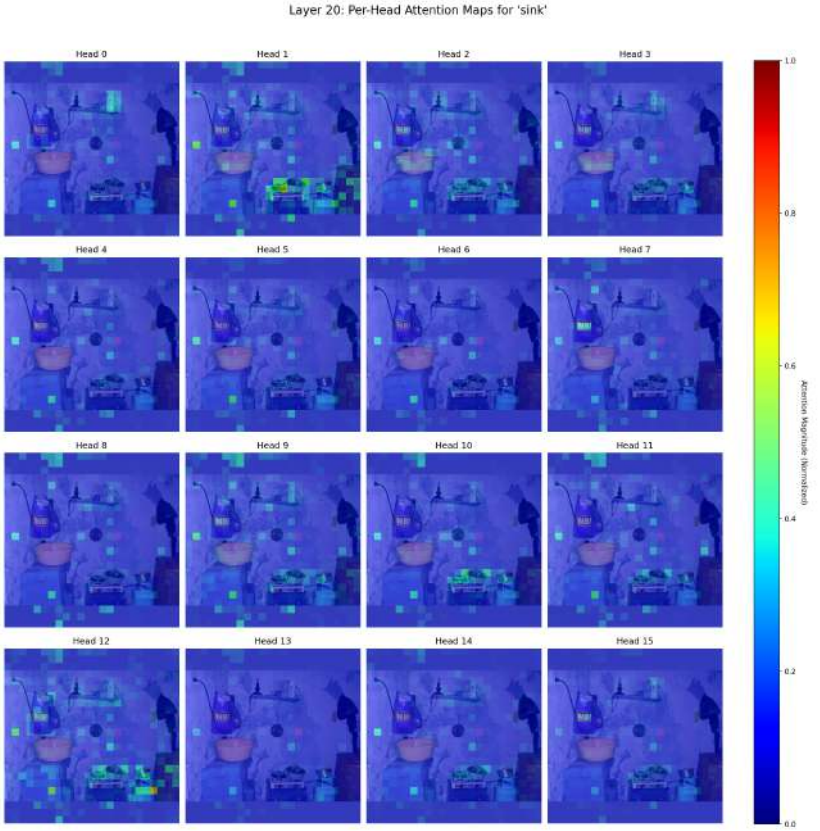


Figure 11: Layer 20 per head attention maps

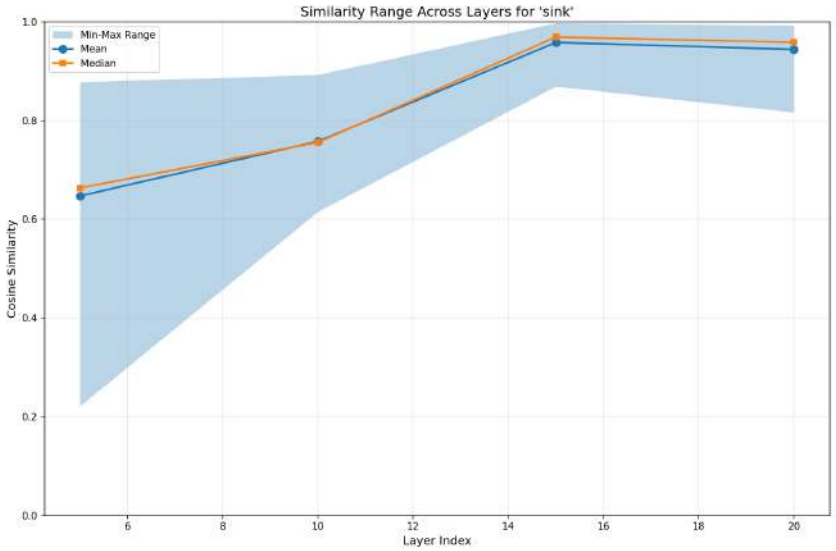


Figure 12: Min-max plot of cosine similarity between heads across layers